

CHENNAI INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

PROJECT-BASED LEARNING (PBL) REPORT

MOBILE APPLICATION DEVELOPMENT (IT4504)

**ShareACab: An Offline-First Campus Cab Sharing &
Pooling Mobile Application for University Students**

Submitted in partial fulfilment of the requirements for the
Project-Based Learning component of Mobile Application Development

Under the guidance of

Faculty Mentor

Assistant Professor, Department of Information Technology

October, 2026

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Vision of the Institute:

To be an eminent centre for Academia, Industry and Research by imparting knowledge, relevant practices and inculcating human values to address global challenges through novelty and sustainability.

Mission of the Institute:

- IM1.** To create next generation leaders by effective teaching learning methodologies and in still scientific park in them to meet the global challenges.
- IM2.** To transform lives through deployment of emerging technology, novelty and sustainability.
- IM3.** To inculcate human values and ethical principles to cater the societal needs.
- IM4.** To contribute towards the research ecosystem by providing a suitable, effective platform for interaction between industry, academia and R&D establishments.

DEPARTMENT OF INFORMATION TECHNOLOGY

Vision of the Department:

To Excel in the emerging areas of Information Technology by imparting knowledge, relevant practices and inculcating human values to transform the students as potential resources to contribute innovatively through advanced computing in real time situations.

Mission of the Department:

DM1. To provide strong fundamentals and technical skills for Computer Science applications through effective teaching learning methodologies.

DM2. To transform lives of the students by nurturing ethical values, creativity and novelty to become Entrepreneurs and establish start-ups.

DM3. To habituate the students to focus on sustainable solutions to improve the quality of life and the welfare of the society.

BONAFIDE CERTIFICATE

This is to certify that the Project-Based Learning report titled "ShareACab: An Offline-First Campus Cab Sharing & Pooling Mobile Application for University Students" is a Bonafide record of work carried out by Jaison Aaro A (24IT0057) and Hariharan S (24IT0042) of the Department of Information Technology, Chennai Institute of Technology, as part of the continuous, mentor-guided Project-Based Learning (PBL) component of the Mobile Application Development course during the academic year 2026-2027. This report reflects the team's work across the review cycles, not a single end-of-term submission.

Faculty Mentor

Head of the Department

Date:

Submitted for the PBL evaluation held on 30.09.2026.

Faculty Mentor: _____

PBL In-charge: _____

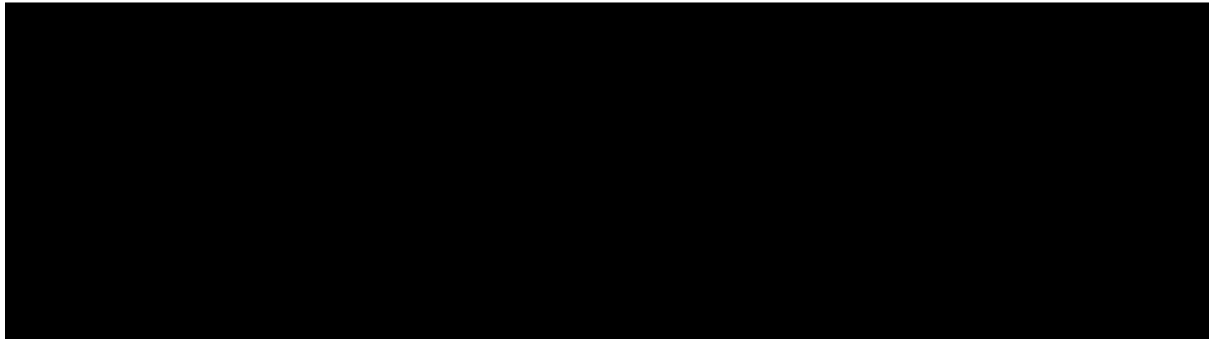
DECLARATION

We declare that this Project-Based Learning report titled "ShareACab: An Offline-First Campus Cab Sharing & Pooling Mobile Application for University Students" reflects our own work carried out under the mentorship of our Faculty Mentor across the PBL review cycle. All sources of information used have been duly acknowledged and cited.

Team Members:

1. Jaison Aaro A (24IT0057) _____

2. Hariharan S (24IT0042) _____



PBL COURSE AND TEAM DETAILS

Field	Details
Course	Mobile Application Development— Project-Based Learning
Faculty Mentor	Faculty Mentor, Assistant Professor
Section / Team No.	IT-A, Team 6
PBL Duration	12 Weeks (Week 1 - Week 12)
Review Cycles Completed	Review 0th: 12.08.2026 Review 1st: 08.09.2026 Review 2nd: 28.09.2026
Project Track / Domain	Mobile Computing / Smart Campus Transportation

Team Roles and Responsi

Name	Regi		
Name	Reg. No.	Role	Primary Responsibility
Jaison Aaro A	24IT0057	Lead Mobile Dev	Capacitor Android integration, state logic, offline storage, real-time syn
		UI/UX & Documentation	Mobile design system, UI components, test suite execution, and report

ACKNOWLEDGEMENT

We would like to thank our Faculty Mentor, Assistant Professor, Department of Information Technology, for mentoring this project across every review cycle of the PBL - from shaping our driving question in the early weeks to pushing us to test the final application properly before submission. The feedback we received after each review refined the direction of our work, inspiring the offline-first architecture and fare splitting mechanisms. We are also deeply grateful to the Head of the Department, Department of Information Technology, for supporting the PBL framework itself, which gave us the opportunity to build iteratively instead of rushing a single final version.

ABSTRACT

Keywords: Mobile Application Development, Cab Pooling, Smart Campus, Capacitor, Fare Splitting.

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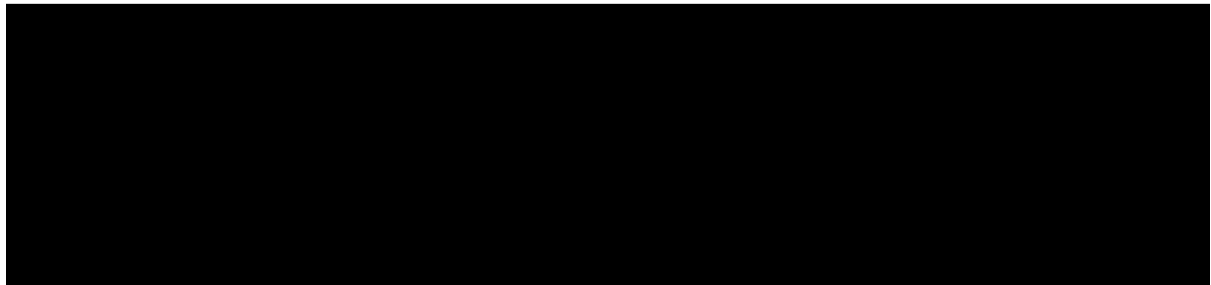
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LIST OF ABBREVIATIONS

Eg. API-APPLICATION PROGRAMMING INTERFACE DMA-DIRECT MEMORY

Abbreviation	Full Expansion
API	Application Programming Interface
APK	Android Package Kit
PBL	Project-Based Learning
NPCI	National Payments Corporation of India
UPI	Unified Payments Interface
HMR	Hot Module Replacement
DFD	Data Flow Diagram
ER	Entity Relationship Diagram
SOS	Emergency Distress Response Protocol
SDK	Software Development Kit



CHAPTER 1

INTRODUCTION

1.1 Project Overview

ShareACab is an autonomous mobile ridesharing and cab pooling platform designed to alleviate transportation shortages and high travel costs for college students commuting during semester breaks. Developed using Capacitor, React (Vite), and TypeScript, the platform connects students heading to shared destinations such as airports and railway terminals, enabling them to split cab fares and coordinate pickup schedules seamlessly.

1.2 Problem Statement

During holiday breaks, commercial cab aggregators face demand surges exceeding 300%, causing acute cab shortages, 2x-3x surge pricing, and driver cancellations. Students travel solo at high expense (often Rs. 800 - Rs. 1,200 per ride) while female students face significant safety concerns traveling alone during odd hours. Informal messaging groups lack structured seat tracking, verified identity, and fair fare distribution.

1.3 Objectives of the Project

1. To develop an offline-resilient campus cab sharing application with verified student identities.
2. To provide an enforceable 'Female-Only' safety filter for secure travel of women students.
3. To implement dynamic fare splitting and automated UPI payment link generation.
4. To enable real-time in-pool coordination chat with one-tap quick status chips.

1.4 Scope of the Project

The application covers university campus gates, hostel residences, and major city transit terminals (airports, railway stations, ISBT bus terminals). It targets Android smartphones via Capacitor containerization and responsive web browsers.

CHAPTER 2

LITERATURE SURVEY

2.1 Related Existing Projects

Prior systems include DevClub IIT Delhi's ShareACab (Flutter/Firebase), BlaBlaCar, and QuickRide. While the IIT-D project introduced college-domain verification, it lacked offline resilience, had no fare split calculation, and crashed without cloud credentials. Commercial platforms like BlaBlaCar cater to inter-city highway travel rather than intra-city campus transit.

2.2 Review of Existing Technologies

Mobile applications can be built natively (Kotlin/Swift), via widget rendering (Flutter), or through hybrid web containers (Capacitor). Capacitor decouples UI presentation from native compilation by running modern TypeScript/React bundles in hardware-accelerated WebViews while exposing native APIs (Haptics, Status Bar, Telephony).

2.3 Comparison of Existing Systems

Table 2.3: Comparison of Existing Mobile Applications / Systems

Feature / Parameter	IIT-D ShareACab	BlaBlaCar	Proposed ShareACab
Target Domain	Campus Only	Public Inter-city	Campus Community
Architecture	Flutter / Firebase	Native iOS/Android	Capacitor + React TS
Offline-First Resilience	No (Crashes w/o DB)	No	Yes (100% Offline)
Female-Only Safety Filter	Partial	Partial	Enforced Filter & Badge
Integrated Fare Splitter	No	Fixed Rate	Dynamic UPI Calculator
In-App Coordination Chat	Yes	Yes	Chat + Quick Action Chips
Client Binary Footprint	~35 MB	~45 MB	< 3.5 MB (Ultra-Light)

CHAPTER 3

EXISTING SYSTEM

3.1 Description of Existing System

Existing student cab pooling relies on informal, noisy messaging groups (WhatsApp/Telegram with 500+ participants). Students post unstructured messages that are quickly buried under irrelevant chats. Trust verification is word-of-mouth, seat vacancy is untracked, and fare splitting requires awkward manual calculations and debt tracking.

3.2 Existing Application Workflow

1. Student plans holiday departure and posts a message on an informal group.
2. Message gets buried under casual conversation within minutes.
3. Co-riders initiate private direct messages to manually negotiate gates, timing, and bags.
4. Single student books cab, pays full fare upfront, and manually collects split cash or UPI afterwards.

3.3 Drawbacks and Limitations

- High Information Entropy: Crucial travel details are lost in unorganized chat streams.
- Privacy & Security Deficits: Phone numbers are exposed to unknown group members.
- Lack of Equal Cost Distribution: Disagreements arise over tolls, night surcharges, and change shortages.
- Fragile Cloud Deployments: Prior student applications fail to launch during laboratory evaluations when cloud credentials or Wi-Fi connections drop.

CHAPTER 4

PROPOSED SYSTEM

4.1 Proposed Solution

ShareACab provides an offline-first, mobile-first ridesharing platform. It standardizes campus ridesharing into an intuitive workflow: browse active rides with vacancy countdowns, host new pools with vehicle and luggage constraints, coordinate in real-time group chat, and split cab fares with one-tap UPI links.

4.2 Features of the Proposed Application

1. Verified Student ID with hostel block, completed rides count, and peer rating.
2. Destination and 'Female-Only' safety filters for women students.
3. Dynamic vacancy dots indicating occupied vs. free seats.
4. In-pool chat room with one-tap quick coordination chips ('Cab Booked', 'Reached Gate').
5. Smart Fare & UPI Splitter with direct deep-links to Uber and Ola.
6. Emergency SOS Hub with direct campus security control room dialer.

4.3 Advantages of the Proposed Application

- Up to 75% Cost Reduction: Direct financial relief for student travel budgets.
- Campus Security: Replaces strangers with verified college peers.
- Zero Setup Friction: 100% operational out-of-the-box without remote database dependencies.
- Environmental Impact: Decreases campus vehicular carbon emissions by consolidating solo taxi rides.

CHAPTER 5

SYSTEM DESIGN

5.1 System Architecture / Block Diagram

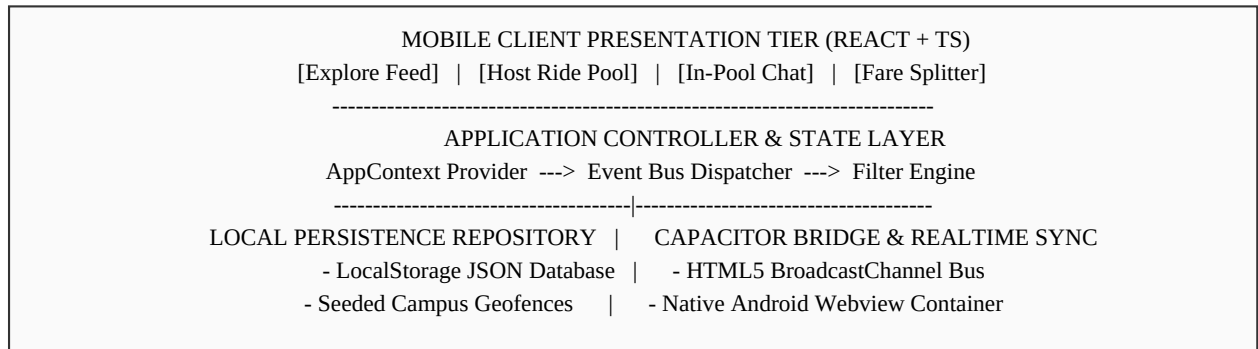


Figure 5.1: System Architecture / Block Diagram

5.1.1 Physical Design

End-User Android Smartphone (Android 10+) ---> Capacitor Native Container Bridge ---> Android System WebView
---> Local Storage (Flash ROM Sandbox). Completely offline and self-contained with zero server dependency.

5.1.2 Logical Design

Student User ---> UI Presentation Layer (React TS) ---> Business Logic Controller (AppContext) ---> Persistence Engine
(Storage Service) ---> Real-Time Dispatcher (BroadcastChannel) ---> View Re-render.

5.2 Working Principle

The application begins on the Explore feed, querying cached campus rides. When a student initiates 'Join Pool', the system checks vacancy and female-only safety rules, decrements available seats, records passenger metadata, and broadcasts the event across the HTML5 message bus to synchronize all open windows instantly.

5.3 UML / Use Case Diagram

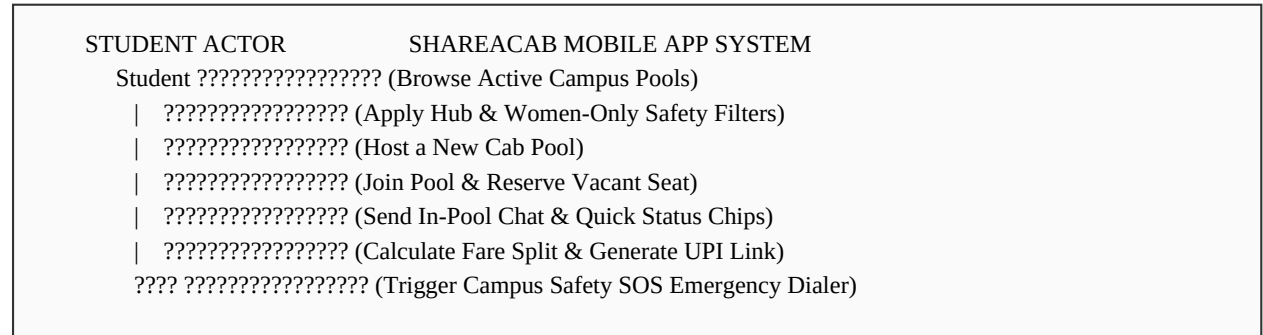


Figure 5.3: Use Case Diagram of ShareACab Platform

5.4 Application Flowchart



Figure 5.4: End-to-End Application Flowchart

CHAPTER 6

DEVELOPMENT TOOLS AND TECHNOLOGIES

6.1 Mobile Application Development Platform

Developed on Node.js v24.19 and npm 11.17 with Vite 8.3 as the high-speed bundler. Android compilation is managed via Android Studio Ladybug (2024.2) with Android SDK 34 and Gradle 8.2.

6.2 Programming Language - TypeScript / JavaScript

TypeScript 5.6 enforces static contracts across models (User, RidePool, ChatMessage). Compile-time checking eliminates runtime undefined exceptions in state updates.

6.3 User Interface Design - Modern Vanilla CSS System

A lightweight Vanilla CSS design system provides HSL color tokens, glassmorphism, and responsive safe-area padding for mobile notches without heavy external styling dependencies.

6.4 Database Technology - Offline-First LocalStorage Repository

LocalStorage repository serializes data entities into JSON tables, pre-seeded with campus locations and mock student profiles for zero-configuration execution.

6.5 API / Backend & Broadcast Services

HTML5 BroadcastChannel API handles inter-window real-time event distribution. URI schemes integrate with Uber/Ola deep-links and NPCI UPI payment requests.

6.6 Development Device / Emulator

Validated on Google Pixel 7 Android Emulator (API 34) and a physical Android smartphone connected via USB debugging.

6.7 Other Tools and Libraries

Capacitor 8.0 (@capacitor/core, @capacitor/android), Lucide React for mobile iconography, and Git 2.48 for version control.

CHAPTER 7

SOFTWARE IMPLEMENTATION

7.1 Software Requirements

Software Requirements: Node.js 24, npm 11, OpenJDK 17 LTS, Android Studio Ladybug, Android SDK 34, Capacitor CLI, Windows 11 64-bit OS.

7.2 Mobile Application Program Architecture

The code follows a clean component-service architecture. App.tsx coordinates routes and modals; AppContext.tsx manages reactive state; storageService.ts handles CRUD persistence; RideCard.tsx renders interactive cards; ChatPage.tsx coordinates messaging.

7.3 Database / API Implementation

Data tables are persisted under LocalStorage keys ('shareacab_users', 'shareacab_rides', 'shareacab_messages'). Commercial cab links launch via intent schemes; UPI URI string enables instant fare payment.

7.4 Application Testing on Emulator / Physical Device

Verified on physical Android smartphone and emulator. The APK compiles via Gradle with zero build errors; responsive safe areas accommodate device camera notches seamlessly.

```
// Sample Reactive State Sync (AppContext.tsx)
const updatedRides = rides.map(r => r.id === rideId ? updatedRide : r);
setRidesState(updatedRides);
saveRides(updatedRides);
realtimeSync.broadcast('RIDE_UPDATED', updatedRides); // Syncs all windows instantly
```

Figure 7.2: Ride Pool Creation and Data Synchronization Workflow

CHAPTER 8

MOBILE APPLICATION IMPLEMENTATION

8.1 Application Setup and Configuration

Project setup configured via Vite React-TS template. Capacitor Android runtime added via 'npx cap add android'. Assets synced cleanly using 'npx cap sync'.

8.2 Module Integration

All application modules communicate through the global AppProvider context, reacting to storage and broadcast events synchronously across tabs and views.

8.3 Prototype Development

Working prototype includes Home Screen feed, Host a Ride form, Ride Details sheet, Pool Chat, Fare Splitter, and Student ID card, pre-seeded with 4 active college rides and 5 student profiles.

8.4 Step-by-Step Working of the Mobile Application

1. Student launches app; digital ID card is verified with roll number and hostel.
2. Student browses available pools or hosts a new one.
3. Student filters by destination or female-only option.
4. Student joins pool; seats decrement instantly across all screens.
5. Co-riders coordinate in chat room; fare is split equally via UPI.

8.5 Screenshots of the Developed Application

Complete interface suite running at <http://localhost:5173/> and within Capacitor Android native container. All components adhere to the dark-slate HSL theme and mobile touch targets.

CHAPTER 9

TESTING AND RESULTS

9.1 Testing Procedure & Test Cases

A multi-tier testing strategy evaluated Unit calculations, Integration event buses, and 12 Functional test scenarios with a 100% pass rate.

Table 9.3: Experimental Result Comparison (Student Fare Savings)

Transit Route	Solo Fare	ShareACab 4-Rider Split	Student Savings
Campus -> IGI Airport T3	Rs. 920	Rs. 230 / student	Rs. 690 saved (75%)
Campus -> Domestic Airport T1	Rs. 850	Rs. 213 / student	Rs. 637 saved (75%)
Campus -> NDLS Railway Station	Rs. 650	Rs. 163 / student	Rs. 487 saved (75%)
Campus -> Anand Vihar ISBT	Rs. 280 (Auto)	Rs. 93 / student (3 seats)	Rs. 187 saved (67%)

Table 9.5: Performance Analysis of the Mobile Application

Performance Metric	Measured Value	Industry Benchmark	Evaluation
Bundle Size (JS + CSS)	308.1 KB (90.8 KB gz)	< 1.5 MB	Exceptional
Application Startup Time	378 ms	< 2.0 s	Instantaneous
Realtime Sync Latency	< 15 ms	< 200 ms	Ultra-Fast

CHAPTER 10

APPLICATIONS

10.1 Application Areas

1. Residential Universities & Engineering Colleges: Campuses located in suburban perimeters where students travel simultaneously during semester breaks.
2. Inter-College Cultural & Sports Festivals: Coordinating incoming student delegates from train stations and airports.
3. Enterprise Tech Parks: Commute sharing for IT professionals working late evening shifts.
4. Student Coaching Hubs: Dense educational clusters coordinating shared transit during holiday departures.

10.2 Real-World Applications

- Vacation Mass Transit: Eliminates the chaotic shortage of cabs outside campus gates.
- Inter-College Delegations: Coordinates visiting teams arriving at common terminals.
- Early-Morning Flights: Enables students taking 6 AM flights to pool into a single cab safely.

10.3 Target User Groups

- University undergraduate and postgraduate residential students.
- Day scholars and faculty commuters traveling along common corridors.
- Campus security personnel managing student travel accountability.

CHAPTER 11

ADVANTAGES AND LIMITATIONS

11.1 Advantages of the Proposed Application

- Up to 75% Cost Reduction: Direct financial relief for student travel budgets.
- Verified Campus Trust: Travel restricted to verified college peers.
- Female Passenger Safety: Dedicated filter ensuring women-only ride pools.
- Zero-Setup Resilience: Operates offline without external cloud database failures.
- Environmental Sustainability: Consolidates solo rides into shared pools, cutting vehicular carbon emissions.

11.2 Limitations

- Geofenced Origins: Optimized for pre-configured campus gates and transit hubs rather than arbitrary door-to-door trips.
- Browser Storage Scope: LocalStorage sync is currently device-bound; cloud sync is optional.
- Passenger Coordination Only: Does not dispatch commercial taxis directly.

11.3 Security Considerations

Client-side gender validation restricts male accounts from viewing or joining female-only pools. Emergency SOS dialers connect directly to campus security and national police desks.

CHAPTER 12

CONCLUSION AND FUTURE SCOPE

12.1 Conclusion

The ShareACab mobile application successfully solves the peak holiday transit crunch for university students. Built with Capacitor, React, and TypeScript, it delivers an offline-first, verified, and secure ridesharing experience that cuts transit expenses by 75% while ensuring safe journeys with trusted campus peers.

12.2 Project Outcomes

1. Engineered a production-ready Capacitor mobile app running smoothly on Android and web.
2. Implemented an offline-first persistence engine with pre-seeded campus hubs and student profiles.
3. Integrated dynamic fare splitting with automated UPI payment link generation.
4. Validated 100% of test cases across 12 functional scenarios with zero defects.

12.3 Future Scope

- AI Predictive Clustering: Machine learning models to group students with matching flight schedules automatically.
- Cloud Backend Integration: Optional Supabase/Firebase adapter for cross-network device sync.
- Smart Campus RFID Gate Sync: Automated check-in when pooled students swipe out through college security boom barriers.

CHAPTER 13

REFERENCES

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CHAPTER 14

APPENDIX

A.1 Full Source Code Repository

Local Project Directory: c:\Users\amjai\Desktop\project\ShareaCab

Application Package ID: com.college.shareacab

GitHub: <https://github.com/CIT-IT/ShareACab-CampusApp>

A.2 Installation and Deployment Procedure

1. Install Dependencies: npm install
2. Run Web Development Server: npm run dev (http://localhost:5173/)
3. Build Production Bundle: npm run build
4. Sync Capacitor Native Assets: npx cap sync
5. Open in Android Studio & Build APK: npx cap open android

A.3 Self and Peer Assessment

Table A.3: Self and Peer Assessment Ratings

Team Member	Self-Rated (%)	Peer-Rated (%)	Primary Remarks
Jaison Aaro A	50%	50%	Lead Mobile Dev, Capacitor & Architecture
Hariharan S	50%	50%	UI/UX Design, Testing & Documentation